DIGT2107: Practice of Software Development

Project Iteration 2: Detailed User Stories, Requirements, and Initial Prototype

Course: DIGT2107 – Fall Term 2026
Instructor: Dr. May Haidar
Due Date: October 11, 2026

1. Introduction

Project Name: VolunTier

Team Number: 3

Team Members: Jana Hussien: 222143515 | Mohammad Jalili: 221243084 | Anwar Bonunagui: 222290266 | Crystal Mumtaz-Shah: 220409777

Document Overview: This document outlines the deliverables for Iteration 2 of the project, focusing on defining detailed user stories, specifying functional and non-functional requirements, and developing an initial prototype. This iteration also emphasizes proper documentation and planning, which sets the stage for subsequent development phases. The goal is to establish a solid foundation by clarifying what the system should do and beginning the implementation with a functional prototype.

2. Functional and Non-Functional Requirements

This section specifies the key behaviors and performance expectations of the software.

2.1. Functional Requirements

List the core functional requirements that the system must fulfill, linking them to user needs and stories.

1. The system shall allow users to log in and access features based on their assigned role: volunteer, coordinator, or administrator. Related user story: User Story 1.
2. The system shall allow volunteers to browse, search, and filter available volunteer opportunities. Related user story: User Story 2.
3. The system shall allow volunteers to register for available opportunities while preventing

duplicate registrations and registrations beyond the capacity limit. Related user story: User Story 3.

4. The system shall allow volunteers to withdraw from an opportunity more than three days before the event's scheduled start time. Related user story: User Story 4.
5. The system shall allow volunteers to view their upcoming registrations, participation history, and completed volunteer hours. Related user story: User Story 5.
6. The system shall allow volunteer coordinators to create, post, edit, and cancel volunteer opportunities. Related user story: User Story 6.
7. The system shall allow volunteer coordinators to view registrations and record volunteer attendance and completed hours. Related user story: User Story 7.
8. The system shall allow organization administrators to manage coordinator access and review program summaries. Related user story: User Story 8.
9. The system shall send registered volunteers an in-app or email notification when an opportunity's date, time, location, or status changes. Related user story: User Story 9.

2.2. Non-Functional Requirements

Define the quality attributes and performance benchmarks for the system.

- **Performance:** Opportunity lists, registration results, and participation records should appear within two seconds under normal use.
- **Usability:** The Java desktop interface shall use clear labels, readable text, consistent navigation, and understandable error messages.
- **Reliability:** The system shall save registration, withdrawal, attendance, and volunteer-hour records accurately without creating duplicate records.
- **Security:** The system shall restrict features according to whether the user is a volunteer, coordinator, or administrator.
- **Scalability:** The database shall support hundreds of users and thousands of participation records.
- **Validation:** The system shall validate required fields, dates, capacity limits, and volunteer hours before saving information.
- **Maintainability:** The application shall use a modular structure so that its interface, business rules, and data storage can be updated separately.

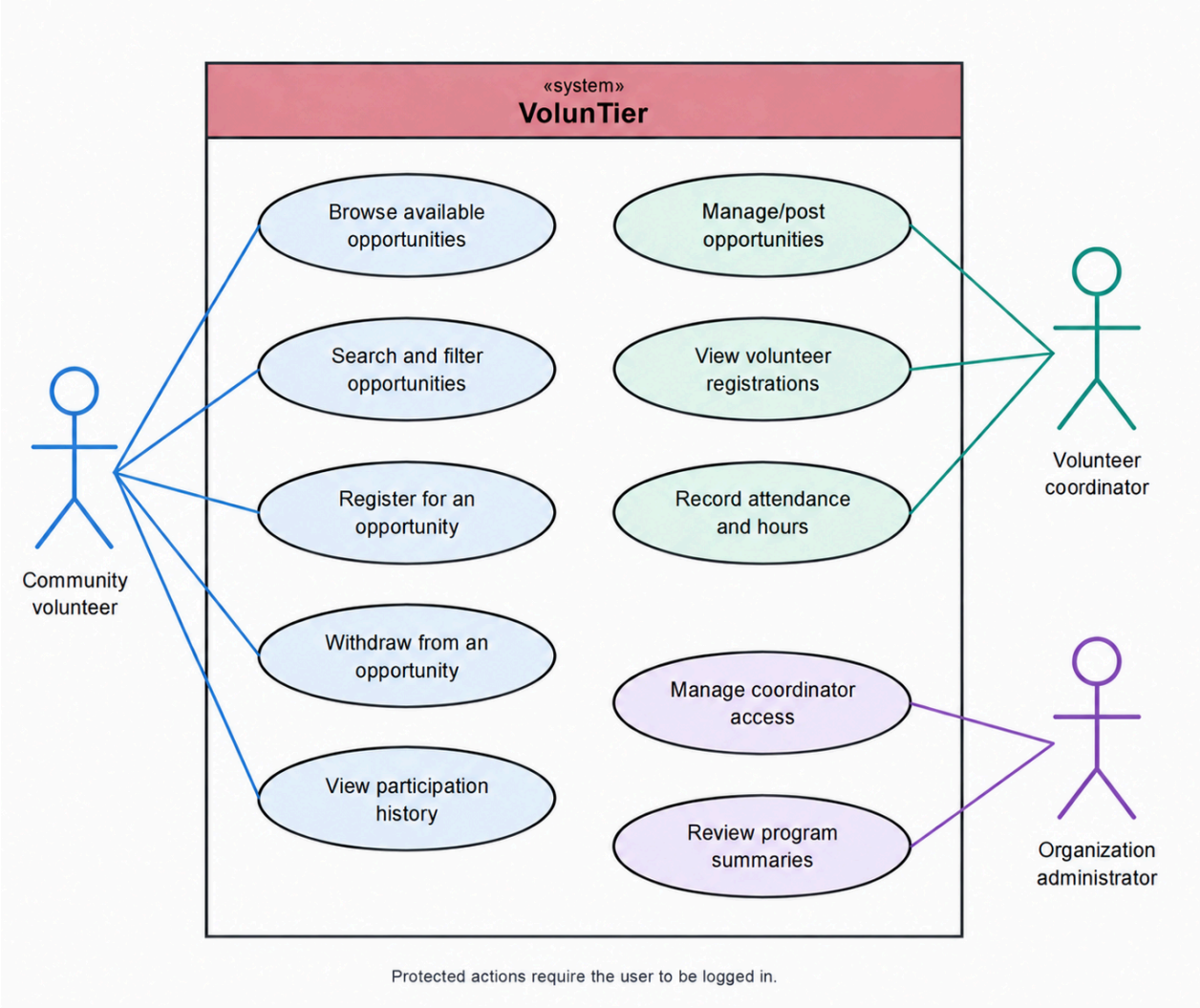
3. Use Cases

Use cases provide a structured approach to understanding the interactions between users and the system.

3.1. Use Case Diagram

Include a UML use case diagram that shows the actors (e.g., Admin, Regular User) and the main use cases (e.g., Manage Inventory, Generate Reports, User Authentication).

The use case diagram identifies the actions available to VolunTier's three user roles. Community volunteers can browse, search, register for, and withdraw from opportunities and view their participation history. Volunteer coordinators can create, post, edit, and cancel opportunities, view registrations, and record attendance and hours. Organization administrators can manage coordinator access and review program summaries. Users must be logged in before accessing features associated with their roles.



3.2. Use Case Descriptions

Detail each key use case with the following structure:

1. Use Case Name: Browse Available Opportunities
   Actors: Community volunteer
   Description: The volunteer views opportunities available through VolunTier.
   Preconditions: The volunteer is logged in.
   Main Flow:
   - The volunteer opens the opportunities page.
   - The system displays available opportunities with their dates, locations, descriptions, and remaining spaces.
   - The volunteer selects an opportunity to view its full details.

Postconditions: The volunteer has viewed an opportunity and can decide whether to register.

2. Use Case Name: Search and Filter Opportunities
   Actors: Community volunteer
   Description: The volunteer narrows the list to find an opportunity that suits them.
   Preconditions: The volunteer is logged in.
   Main Flow:
   - The volunteer enters a search term or selects a filter.
   - The system searches the available opportunities.
   - The system displays the matching opportunities.

Alternative Flow: If there are no matches, the system displays a message and lets the volunteer change the search or filters.
Postconditions: The volunteer sees opportunities matching the selected criteria.

3. Use Case Name: Register for an Opportunity
   Actors: Community volunteer
   Description: The volunteer reserves a place in an opportunity.
   Preconditions: The volunteer is logged in and the opportunity is accepting registrations.
   Main Flow:
   - The volunteer opens an opportunity and selects Register.
   - The system checks that space is available.
   - The system checks that the volunteer has not already registered.
   - The system saves the registration and displays a confirmation.

Alternative Flow: If the opportunity is full or the volunteer is already registered, the system explains why registration cannot be completed.
Postconditions: The opportunity appears in the volunteer's upcoming registrations, and one fewer place is available.

4. Use Case Name: Withdraw from an Opportunity
   Actors: Community volunteer
   Description: The volunteer cancels a registration when they can no longer attend.
   Preconditions: The volunteer is logged in and registered for the opportunity.
   Main Flow:
   - The volunteer opens their upcoming registrations.
   - The volunteer selects an opportunity and chooses Withdraw.
   - The system checks that the event is more than three days away.
   - The system removes the registration and confirms the withdrawal.

Alternative Flow: If the event is within three days, the system prevents the withdrawal and explains the cutoff.
Postconditions: The registration is removed, and the place becomes available to another volunteer.

5. Use Case Name: View Participation History
   Actors: Community volunteer
   Description: The volunteer reviews past participation and completed volunteer hours.
   Preconditions: The volunteer is logged in.
   Main Flow:
   - The volunteer opens their participation history.
   - The system retrieves their past opportunities, attendance records, and recorded hours.
   - The system displays the information to the volunteer.

Alternative Flow: If the volunteer has no records, the system displays a clear message.
Postconditions: The volunteer can review their history without changing any records.

6. Use Case Name: Manage and Post Opportunities
   Actors: Volunteer coordinator
   Description: The coordinator creates, posts, edits, or cancels volunteer opportunities.
   Preconditions: The coordinator is logged in and has coordinator access.
   Main Flow:
   - The coordinator opens the opportunity management page.
   - The coordinator creates an opportunity or selects an existing one to edit.

- The coordinator enters or updates its title, description, date, time, location, and capacity.
- The system validates the information and saves the changes.
- The system updates the opportunity list shown to volunteers.

Alternative Flow: If required information is missing or invalid, the system explains the problem and does not save the change. If the coordinator changes or cancels an existing opportunity, the system notifies registered volunteers.
Postconditions: The opportunity list displays the current information.

7. Use Case Name: View Volunteer Registrations
   Actors: Volunteer coordinator
   Description: The coordinator checks who has registered for an opportunity.
   Preconditions: The coordinator is logged in and has access to the opportunity.
   Main Flow:
   - The coordinator selects an opportunity.
   - The coordinator opens its registration list.
   - The system displays the registered volunteers and the number of available places.

Postconditions: The coordinator has reviewed the registrations without changing them.

8. Use Case Name: Record Attendance and Hours
   Actors: Volunteer coordinator
   Description: The coordinator records which volunteers attended and how many hours they completed.
   Preconditions: The coordinator is logged in, and the opportunity has taken place.
   Main Flow:
   - The coordinator opens the opportunity's registration list.
   - The coordinator records which volunteers attended.
   - The coordinator enters the hours completed by each attending volunteer.
   - The system validates and saves the attendance and hour records.

Alternative Flow: If an hour entry is missing or invalid, the system asks the coordinator to correct it before saving.
Postconditions: Attendance and hours are stored and appear in the relevant volunteers' participation histories.

9. Use Case Name: Manage Coordinator Access
   Actors: Organization administrator
   Description: The administrator controls which users have coordinator permissions.

Preconditions: The administrator is logged in.
Main Flow:

- The administrator opens the user management page.
- The administrator selects a user.
- The administrator grants or removes coordinator access.
- The system saves the change and updates the user's permissions.

Postconditions: The user can access the features permitted by their updated role.

10. Use Case Name: Review Program Summaries
    Actors: Organization administrator
    Description: The administrator reviews an overview of volunteer program activity.
    Preconditions: The administrator is logged in.
    Main Flow:
    - The administrator opens the program summaries page.
    - The system gathers information about opportunities, registrations, attendance, and volunteer hours.
    - The system displays the summary.

Alternative Flow: If there is no information for the selected period, the system displays a clear message.
Postconditions: The administrator has reviewed the program's activity without changing any records.

4. Detailed User Stories

Category 1: Access and Opportunity Discovery

User Story 1: Role-Based Login
As a VolunTier user, I want to log in and access the features assigned to my role so that I can use the system securely.
Acceptance criteria: Valid login information opens the appropriate dashboard. Invalid information displays a clear error. Volunteers, coordinators, and administrators can access only their assigned features.
Priority: High

Story points: 5
Related requirement: 2.1, Requirement 1

User Story 2: Browse and Search Opportunities
As a volunteer, I want to browse, search, and filter opportunities so that I can find one that fits my interests and availability.
Acceptance criteria: Available opportunities display their main details. Search and filters narrow the results. The system displays a message when there are no matches.
Priority: High
Story points: 3
Related requirement: 2.1, Requirement 2

Category 2: Registration and Participation

User Story 3: Register for an Opportunity
As a volunteer, I want to register for an opportunity so that my place is reserved.
Acceptance criteria: A successful registration appears in my upcoming opportunities. I cannot register twice for the same opportunity. Registration is prevented when capacity is reached, with a clear explanation.
Priority: High
Story points: 5
Related requirement: 2.1, Requirement 3

User Story 4: Withdraw from an Opportunity
As a volunteer, I want to withdraw when I can no longer attend so that my place becomes available to someone else.
Acceptance criteria: I can withdraw more than three days before the event. My registration is removed and the place becomes available again. Withdrawal within three days is prevented with an explanation.
Priority: Medium
Story points: 3
Related requirement: 2.1, Requirement 4

User Story 5: View My Registrations and Hours
As a volunteer, I want to see my upcoming registrations, participation history, and completed hours so that I can keep track of my volunteer work.
Acceptance criteria: Upcoming and past opportunities are displayed clearly. Completed hours appear after a coordinator records them. A helpful message appears if I have no participation records.

Priority: Medium
Story points: 3
Related requirement: 2.1, Requirement 5

Category 3: Coordinator Tools

User Story 6: Manage Opportunities
As a volunteer coordinator, I want to create, post, edit, and cancel opportunities so that volunteers have accurate information about events.
Acceptance criteria: I can enter an opportunity's title, description, date, time, location, and capacity. Missing or invalid information is rejected with a clear message. Saved changes appear in the opportunity list.
Priority: High
Story points: 5
Related requirement: 2.1, Requirement 6

User Story 7: Track Attendance and Hours
As a volunteer coordinator, I want to view registrations and record attendance and completed hours so that volunteer participation is documented accurately.
Acceptance criteria: I can view the registered volunteers for an opportunity. I can save attendance and valid hour amounts. Saved hours appear in each volunteer's participation history.
Priority: Medium
Story points: 5
Related requirement: 2.1, Requirement 7

Category 4: Administration and Notifications

User Story 8: Oversee the Volunteer Program
As an organization administrator, I want to manage coordinator access and review program summaries so that I can oversee the volunteer program.
Acceptance criteria: I can grant or remove coordinator access. The user's available features reflect the access change. I can review a summary of opportunities, participation, and volunteer hours.
Priority: Medium
Story points: 5
Related requirement: 2.1, Requirement 8

User Story 9: Receive Opportunity Updates
As a registered volunteer, I want to know when an opportunity changes or is cancelled so that I

can adjust my plans.
Acceptance criteria: A change to the date, time, location, or status triggers a notification to registered volunteers. The notification identifies the opportunity and explains what changed. It is delivered in-app or by email, depending on the notification method the team implements.
Priority: Medium
Story points: 5
Related requirement: 2.1, Requirement 9

5.1. Prototype Features

Seven wireframes using mock data and user flows

- Browse Opportunities
- Sign In
- Opportunity Details
- My Volunteering
- Post an Opportunity
- Attendance & Hours
- Program Overview

5.2. Development Approach

We began with the main screens needed for volunteers to find and register for opportunities, then added the coordinator and administrator screens. We kept the screens separate so they can be reviewed and changed as the requirements are refined. The wireframes currently use mock data. In the next iteration, we plan to build the working Java application, connect the screens, add data storage, and test the main workflows.

5.3. Task Completion and Team Collaboration

Task Estimates

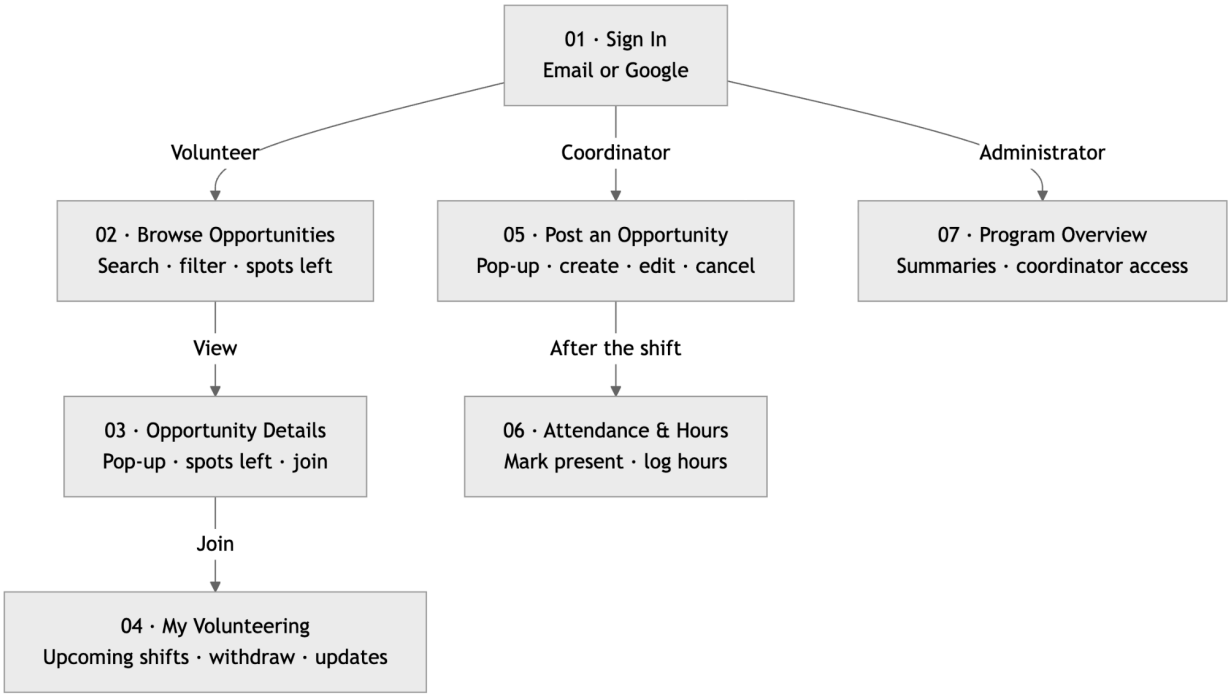
- Setup and mock data: 7 hours
- Sign In: 4 hours
- Browse: 5 hours
- Details and Register: 5 hours
- My Volunteering: 5 hours

- Post an Opportunity: 5 hours
- Attendance & Hours: 4 hours
- Program Overview: 4 hours
- Notifications: 3 hours
- Testing: 3 hours

Total Estimated Time: 45 hours

These are estimates for building and testing the working prototype, not hours already completed. For this iteration, the wireframes and user flows have been prepared. The next development tasks are to implement the screens and navigation, connect them to the application logic and data, and test actions such as registration and attendance recording. We will track those tasks and update their status as development continues.

Tasks are tracked as GitHub issues, and we commit often.
Prototype screenshots are also in the Github Repo
Added Github Issues



VolunTier

Show up for your community.

Find volunteer shifts near you, reserve your spot in one click, and keep track of every hour you give.

Sign in

Continue with Google

or sign in with email

Email

crystal.ms@example.com

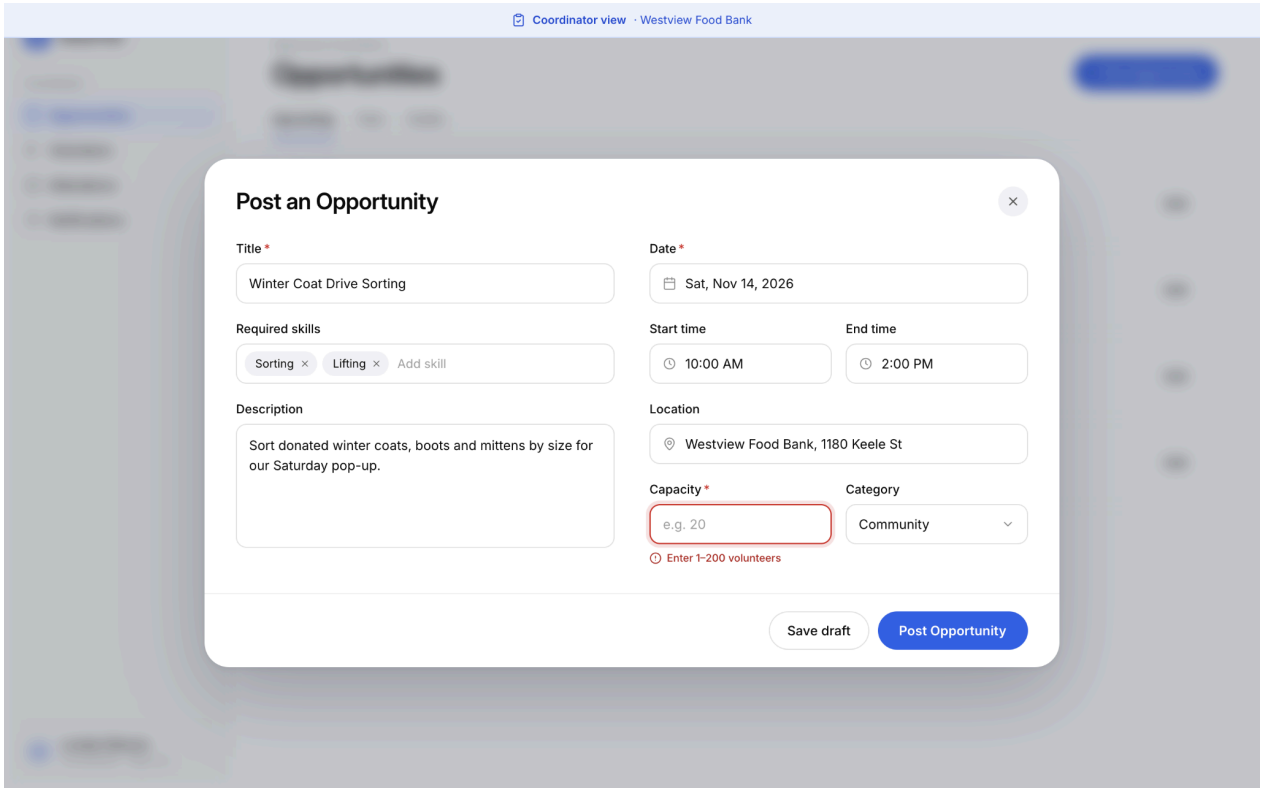
Password Forgot password?

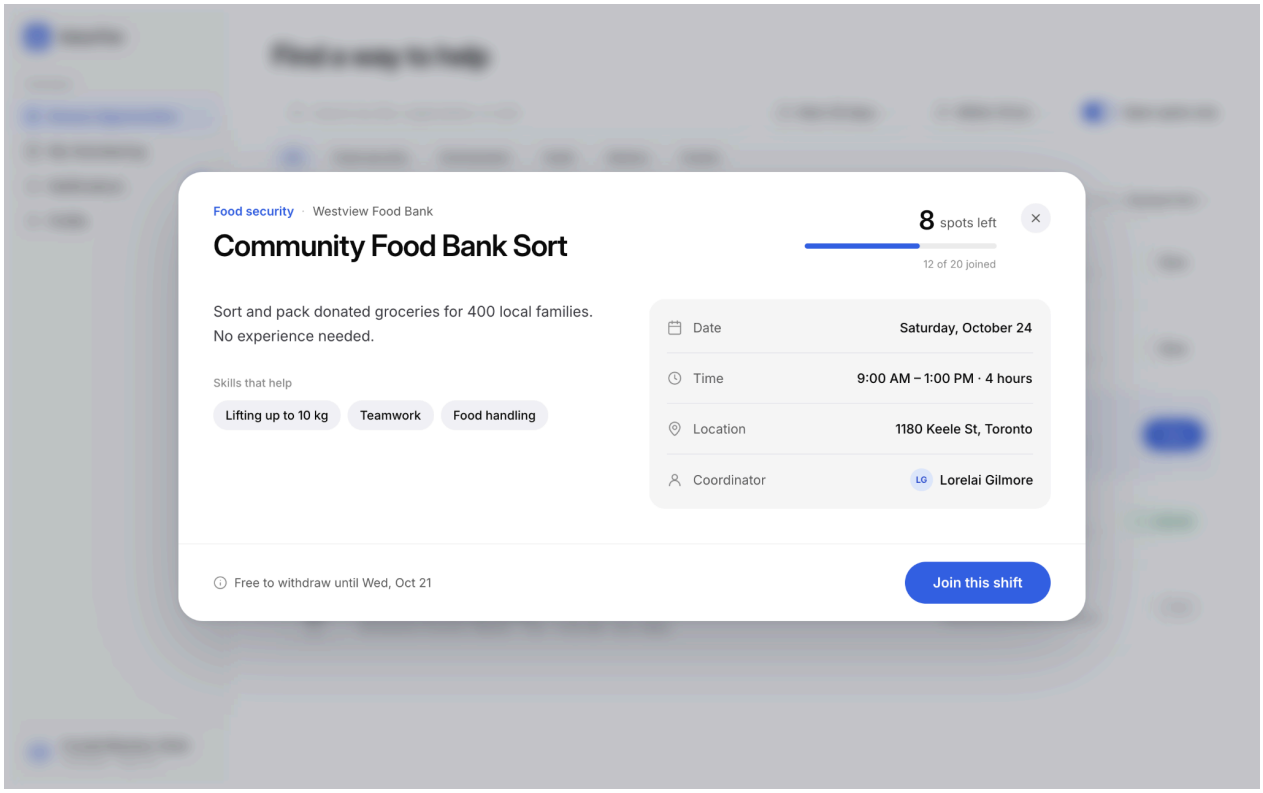
••••••••••

Sign in

Create an account

Terms of use Privacy policy Help





VolunTier

Volunteer

- Browse Opportunities
- My Volunteering
- Notifications 2
- Profile

CM Crystal Mumtaz-Shah
Volunteer · Sign out

You're in! Community Food Bank Sort was added to your upcoming shifts. 7 of 20 spots left

My Volunteering

Upcoming

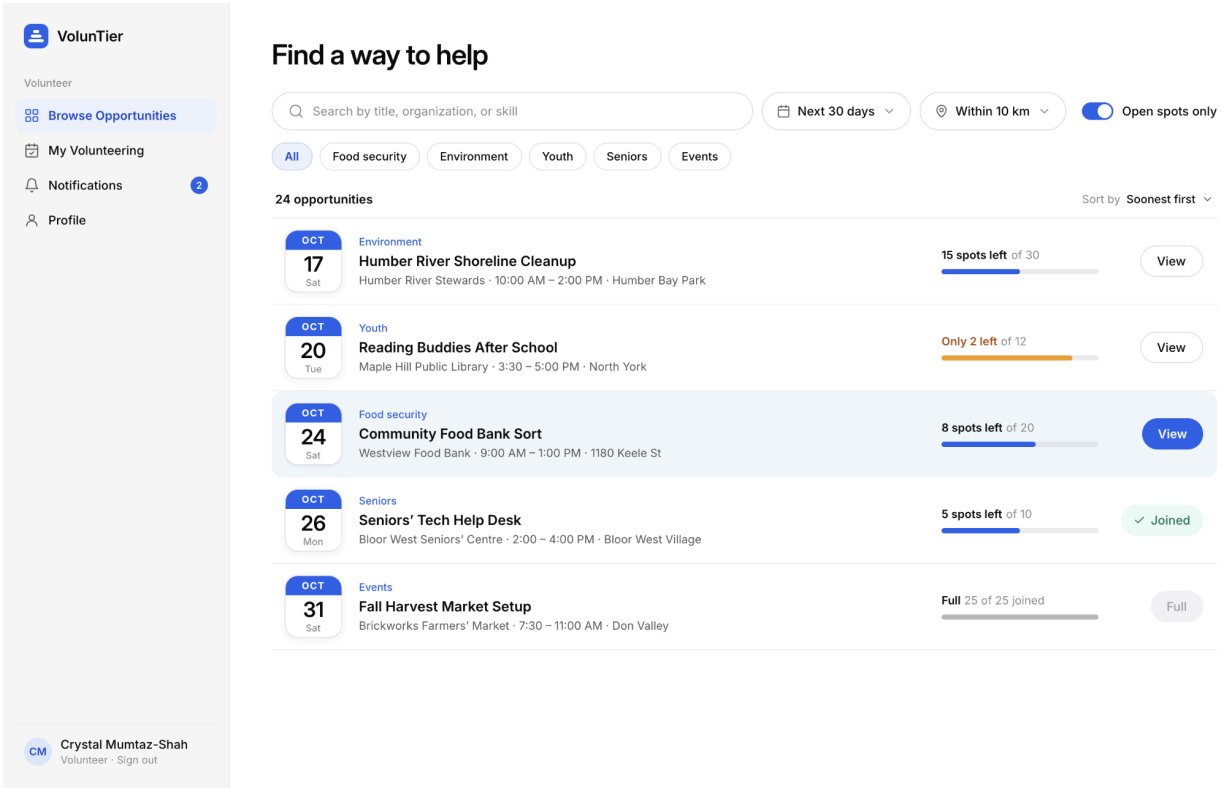
Withdraw up to 3 days before a shift

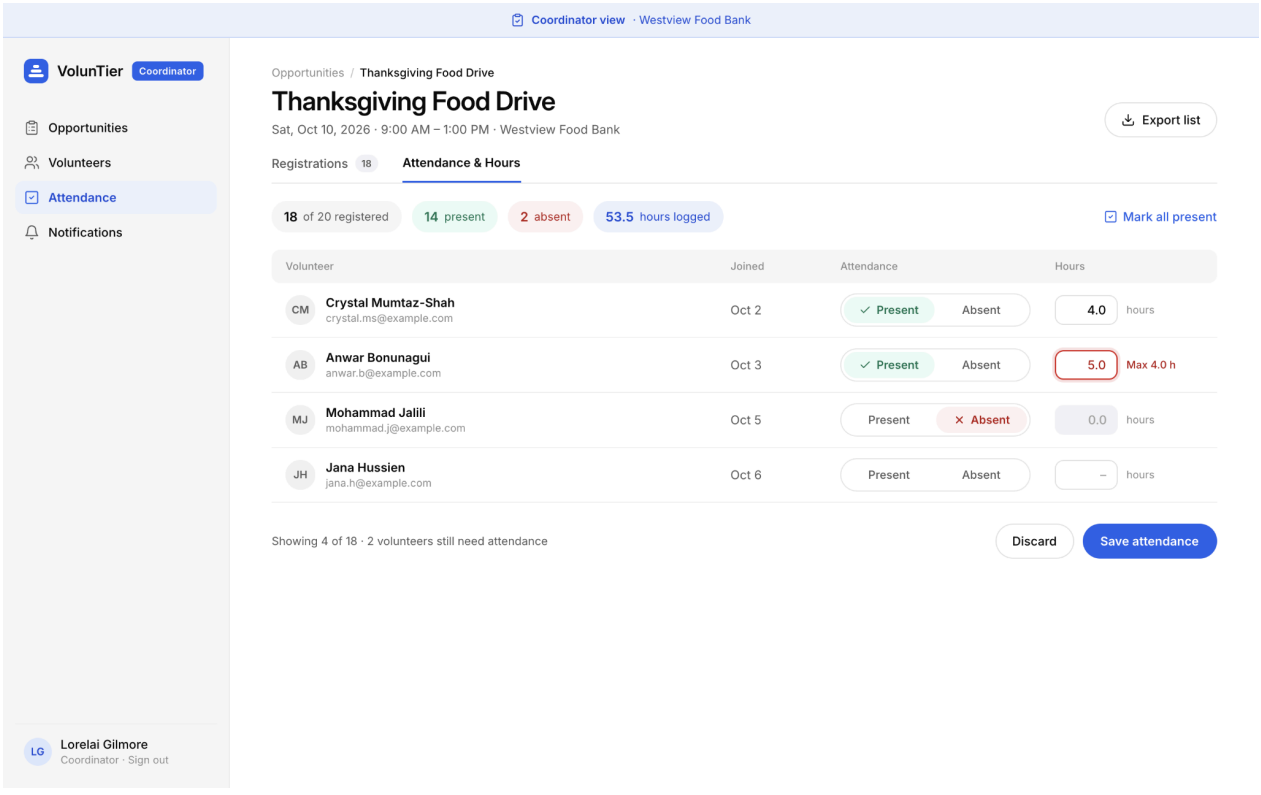
OCT 17 Sat	**Humber River Shoreline Cleanup** Humber River Stewards · 10:00 AM – 2:00 PM	Withdrawal closed Shift is in 2 days
OCT 24 Sat	**Community Food Bank Sort** New Westview Food Bank · 9:00 AM – 1:00 PM	Withdraw until Wed, Oct 21
OCT 26 Mon	**Seniors' Tech Help Desk** Bloor West Seniors' Centre · 2:00 – 4:00 PM Time changed from 1:00 PM · Oct 14	Withdraw until Fri, Oct 23

Updates

View all

- **Seniors' Tech Help Desk now starts at 2:00 PM** — Oct 14
  Bloor West Seniors' Centre changed the time. Your spot is still reserved.
- **Reminder: Shoreline Cleanup is this Saturday** — Today
  Bring gloves, water and closed-toe shoes. Meet at the Humber Bay Park east lot.


VolunTier
Volunteer
Browse Opportunities
My Volunteering
Notifications 2
Profile
Find a way to help
Search by title, organization, or skill
Next 30 days
Within 10 km
Open spots only
All
Food security
Environment
Youth
Seniors
Events
24 opportunities
Sort by Soonest first
OCT 17 Sat
Environment
Humber River Shoreline Cleanup
Humber River Stewards · 10:00 AM – 2:00 PM · Humber Bay Park
15 spots left of 30
View
OCT 20 Tue
Youth
Reading Buddies After School
Maple Hill Public Library · 3:30 – 5:00 PM · North York
Only 2 left of 12
View
OCT 24 Sat
Food security
Community Food Bank Sort
Westview Food Bank · 9:00 AM – 1:00 PM · 1180 Keele St
8 spots left of 20
View
OCT 26 Mon
Seniors
Seniors' Tech Help Desk
Bloor West Seniors' Centre · 2:00 – 4:00 PM · Bloor West Village
5 spots left of 10
Joined
OCT 31 Sat
Events
Fall Harvest Market Setup
Brickworks Farmers' Market · 7:30 – 11:00 AM · Don Valley
Full 25 of 25 joined
Full
CM
Crystal Mumtaz-Shah
Volunteer · Sign out


Coordinator view · Westview Food Bank
VolunTier Coordinator
Opportunities
Volunteers
Attendance
Notifications
Opportunities / Thanksgiving Food Drive
Thanksgiving Food Drive
Sat, Oct 10, 2026 · 9:00 AM – 1:00 PM · Westview Food Bank
Export list
Registrations 18
Attendance & Hours
18 of 20 registered
14 present
2 absent
53.5 hours logged
Mark all present
Volunteer
Joined
Attendance
Hours
CM Crystal Mumtaz-Shah crystal.ms@example.com Oct 2 Present Absent 4.0 hours
AB Anwar Bonunagui anwar.b@example.com Oct 3 Present Absent 5.0 Max 4.0 h
MJ Mohammad Jalili mohammad.j@example.com Oct 5 Present Absent 0.0 hours
JH Jana Hussien jana.h@example.com Oct 6 Present Absent – hours
Showing 4 of 18 · 2 volunteers still need attendance
Discard
Save attendance
LG
Lorelai Gilmore
Coordinator · Sign out

Admin view · You're managing the whole program. Changes here apply to every coordinator.

VolunTier Admin

- Program Overview
- Coordinators
- Reports
- Settings

EG Emily Gilmore
Administrator · Sign out

Fall 2026 · Sep 1 – Oct 15

Program Overview

Fall 2026 | Download summary

Volunteer hours	Active volunteers	Opportunities posted	Attendance rate
1,284	312	46	91%
+12% vs. summer term	+38 new this term	14 still upcoming	Above the 85% goal

Hours by program

Sep 1 – Oct 15

Food security	486
Environment	322
Youth	241
Seniors	158
Events	77

Food security drew 38% of all hours this term.

Coordinator access

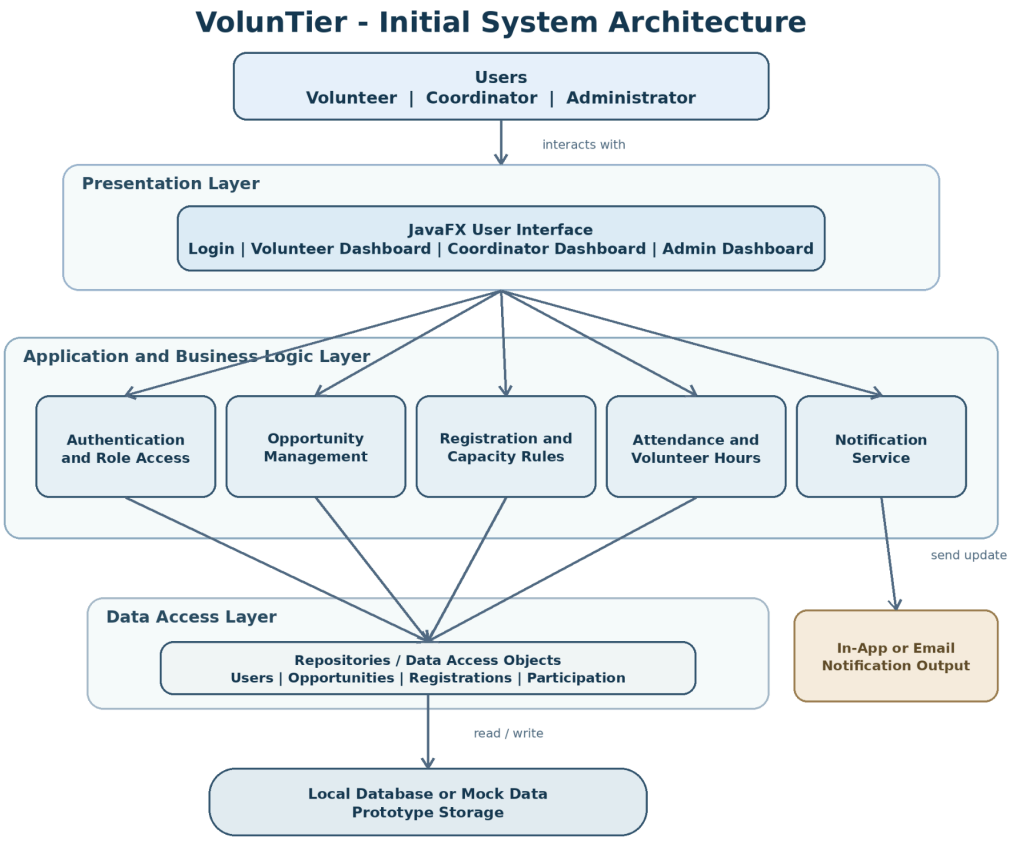
+ Invite coordinator

LG	Lorelai Gilmore Food security · 18 shifts	Active
LD	Luke Danes Environment · 12 shifts	Active
RG	Rory Gilmore Youth programs · 9 shifts	Active
KG	Kirk Gleason Events · 7 shifts	Paused

Paused coordinators keep their history but can't post or edit opportunities.

6. Documentation Updates

6.1. System Architecture Documentation



VolunTier uses a layered structure that separates the user interface, application logic, and data storage. This structure makes the system easier to understand, test, and update as development continues. The presentation layer contains the JavaFX user interface. It provides separate screens and dashboards for volunteers, volunteer coordinators, and organization administrators.

Volunteers can search for opportunities, sign up, cancel, and review their volunteer history. Coordinators can create and edit opportunities, view registered volunteers, and track attendance

and hours. Administrators can manage coordinators and view reports related to the program.

The application and business logic layer contains the main procedures and rules for the system. It includes the authentication and access control mechanism for the three user categories, management of opportunities, registration processes, capacity control, attendance, recording of hours, and notifications. For instance, when volunteers sign up for an opportunity, a check within the registration service will confirm if the slot is available and whether an applicant has not already registered for the event.

The data access layer consists of the application's relational database. It stores the main objects such as users, opportunities, attendance, and volunteers' hours. In the initial prototype, a mock-up or local database can be used. This description will be replaced with an actual version once the prototype implementation will be completed.

The notifications service is responsible for sending messages to volunteers regarding changes and cancellations of opportunities. At first, notifications can be displayed within the application's interface, but the option to send e-mails to volunteers can be considered as well.

6.2. Planning Document and Log File

Original plan from Deliverable 1.1

Iteration 1 (Weeks 1–2, due September 20): Complete the project vision, high-level features, initial user stories, and GitHub repository setup.

Iteration 2 (Weeks 3–5, due October 11): Expand the user stories, write functional and non-functional requirements, create use case diagrams and descriptions, and update the backlog.

Iteration 3 (Weeks 6–9, due November 8): Build an initial working prototype focused on browsing opportunities and registration. Add JUnit tests and prepare a testing plan and report.

Iteration 4 (Weeks 10–12, due November 29): Review another team's prototype, use feedback to improve our work, and submit the updated documentation, user manual, and prototype.

Modified plan for Deliverable 1.2

Iteration 1 remains complete as planned. During Iteration 2, we have defined more detailed requirements, use cases, and user stories; prioritized the backlog; documented the system architecture; and added wireframes with mock data to show the main user workflows. During

Iteration 3, we plan to turn the priority workflows into a working Java desktop prototype, test rules such as capacity limits and duplicate registration, and document the test results. Iteration 4 remains focused on review, feedback, and improvement.

Change log and reasons

1. We added wireframes during Iteration 2. The original 1.1 plan placed the working prototype in Iteration 3. The wireframes let us show and review the planned screens for Deliverable 1.2 before implementing them. The working prototype and its tests remain planned for Iteration 3.
2. We expanded the five initial user stories into detailed requirements, use cases, and a prioritized backlog. This gives the team clearer tasks and helps connect each planned feature to a user need.
3. We set a development order for Iteration 3. Browsing opportunities, registration, role-based access, and coordinator opportunity management are high priorities. Other features will follow according to the backlog and available time.

7. Iteration Plan and Backlog

7.1. Iteration Plan

Iteration 2 spans Weeks 3 to 5. It entails developing the initial ideas described in the VolunTier vision into requirements, user stories, the use case description, and the initial prototype. We will use the feedback received and updated backlog in Section 7.2 to prioritize development tasks.

Week 3: We developed the functional and non-functional requirements and identified key workflows for all three types of users (volunteers, coordinators, and admins). All functional requirements have been linked to their respective user stories in Section 4.

Week 4: Jana added the use case diagram in Section 3.1, while Mohammad described all the use cases in Section 3.2 and updated the user stories in Section 4. We will also make sure that all requirements, including the stories and diagrams, are using the same terms and roles.

Week 5: Crystal will finish developing the initial prototype that will showcase the primary features of the system. She will describe it in detail in Section 5, including the overall approach, features to be implemented, and tasks to be done for each story. Mohammad will finalize the system architecture by providing the architecture diagram and its description in Section 6.1. Anwar will complete the initial and updated plans and explain his rationale for changes in Section 6.2. We will update the project document based on the changes and finalize it during this week. We will update the project's backlog reflecting our progress and document prototype code and

related materials in the GitHub repository, tagging the commit ITR1.2. Finally, we will submit our findings in one PDF file to eClass.

Iteration 3 encompasses Weeks 6 to 9 and focuses on implementing and testing the system. In Week 6, we will analyze the feedback received for the initial prototype and prioritize the backlog items. We will begin implementing opportunity browsing and registration functions and coordinator's opportunity management feature.

In Week 7, we will continue developing these features and introduce unit tests for all the key rules, including the capacity limits and duplicate registration checks.

During Week 8, we will connect all the features into a single system and test the entire workflow for volunteers and coordinators.

Finally, in Week 9, we will address the issues discovered during the tests, finalize the prototype, and finish elaborating the test plan and report. The backlog will be updated based on the feedback and test results.

7.2. Updated Backlog

The priorities and estimates below are initial planning estimates.

1. Browse and search volunteer opportunities
    - Priority: High | Estimate: 3 points | Status: Requirements drafted
2. Register for an opportunity with duplicate-registration and capacity checks
    - Priority: High | Estimate: 5 points | Status: Requirements drafted
3. Allow coordinators to create and edit opportunities
    - Priority: High | Estimate: 5 points | Status: Requirements drafted
4. Provide role-based login for volunteers, coordinators, and administrators
    - Priority: High | Estimate: 5 points | Status: Requirements drafted
5. Allow volunteers to withdraw before the three-day cutoff
    - Priority: Medium | Estimate: 3 points | Status: Requirements drafted
6. Let coordinators record attendance and volunteer hours
    - Priority: Medium | Estimate: 5 points | Status: Requirements drafted
7. Let volunteers view upcoming registrations, history, and completed hours
    - Priority: Medium | Estimate: 3 points | Status: Requirements drafted
8. Notify registered volunteers when an opportunity changes or is cancelled

    - Priority: Medium | Estimate: 5 points | Status: Requirements drafted
9. Let administrators manage coordinator access and view program summaries
    - Priority: Medium | Estimate: 5 points | Status: Requirements drafted
10. Add volunteer profile pictures
    - Priority: Low | Estimate: 2 points | Status: Future consideration
11. Send SMS notifications to volunteers
    - Priority: Low | Estimate: 5 points | Status: Future consideration
12. Generate a downloadable volunteer-hours report
    - Priority: Low | Estimate: 5 points | Status: Future consideration

8. Submission Guidelines

- **GitHub Repository**: Update the repository with the prototype code and all documentation. Tag this version as “ITR1.2”.
- **eClass Submission**: Submit a single PDF file containing:
    - Functional and Non-Functional Requirements
    - Use Case Diagram and Descriptions
    - Detailed User Stories
    - Initial Prototype Overview
    - Updated Iteration Plan and Backlog